

Chapter 4

Quantum Features and Signatures of Quantum Thermal Machines



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The aim of this book chapter is to indicate how quantum phenomena are affecting the operation of microscopic thermal machines, such as engines and refrigerators. As converting heat to work is one of the fundamental concerns in thermodynamics, the platform of quantum thermal machines sheds light on thermodynamics in the quantum regime. This chapter focuses on the basic features of quantum mechanics, such as energy quantization, the uncertainty principle, quantum coherence and correlations, and their manifestation in microscopic thermal devices. In addition to indicating the peculiar behaviors of thermal-machines due to their non-classical aspects, we present quantum-thermodynamic signatures of these machines. Any violation of the classical thermodynamic bounds observed in the outcome of measurements on these machines is a sufficient condition to conclude that quantum effects are present in the operation of that thermal machine. Experimental setups demonstrating some of these results are also presented.

4.1 Introduction

Quantum mechanics is one of the greatest revolutions in the history of science. It changed the way we understand the world and demonstrates that the microscopic

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realm is governed by a theory that is fundamentally different from classical mechanics. Classical mechanics is formulated on the phase-space and treats particles as points whereas quantum mechanics is described by wave functions and operators acting in Hilbert space. In the quantum framework, the classical perception of certainty is replaced by probability, and instead of being continuous, physical quantities are generally quantized. These significant differences result in peculiar phenomena that are observed only in the quantum regime.

Since the beginning, the development of quantum mechanics has been influenced by thermodynamics. Planck initiated the quantum era by introducing the quantization hypothesis to describe thermal radiation emitted from a black body; Einstein discovered stimulated emission while studying thermal equilibration between light and matter. In spite of the close historical relationship between quantum mechanics and thermodynamics, one could wonder if the quantum revolution would eventually shake the foundations of thermodynamics as it did with classical mechanics. Some of the early works in this direction studied the behavior of heat machines, which, besides their technological applications, were used to test the compliance with the laws of thermodynamics. Despite the effort focused on this direction, as of today, the fundamental laws and bounds of thermodynamics hold also in the quantum regime.

Even though quantum mechanics complies with the laws of thermodynamics, classical and quantum heat machines still differ in a non trivial way. As we show in this chapter, not only are classical and quantum heat machines fundamentally different, but quantum mechanics allows the realization of classically inconceivable heat machines.

In order to identify the appearance and effects of quantum features in thermal machines, one should first have a clear notion of classical thermal machines. Throughout this chapter, we will use different notions or levels of classicality, starting from the purest definitions, (i.e., a system is classical only if it is precisely described by classical mechanics), and working toward less strict definitions, which consider systems governed by quantum Hamiltonians as classical as long as either their coherences or quantum correlations are zero. In this manner we obtain a thorough understanding of how different quantum features influence thermal devices.

The chapter is divided into three sections concerning three main quantum features and their corresponding thermodynamic signatures. In the first section, classicality will be considered as it is in classical thermodynamics, where the probability distribution is fully described by a phase-space distribution without any constraint aside from normalization. In this section, quantum effects result only from the quantization of energy levels and the uncertainty principle, which set some limitations on the probability distributions. These features alone lead to discrepancies in the behavior of classical and quantum heat machines.

The second section describes the effects of quantum coherence in thermal machines. Both the positive and negative implications of coherence are discussed. Furthermore, we present a recent experiment demonstrating some of these results. In these scenarios thermal machines that are governed by a quantum Hamiltonian, but can be fully described by their populations are considered classical (stochastic). In this context it's also important to note that the preferred basis to describe thermal

machine is the one in which measurements are preformed, typically, this will be the energy basis.

In the last section, we discuss the role of correlations in the operation of thermal machines. We demonstrate that quantum correlations can induce anomalous heat flow from a colder body to a hotter body, that can not be explained classically. Here, classicality refers to separable quantum states with zero discord, (i.e., only classical correlations are allowed). The chapter is structured so that each of the three main section are independent of the others.

4.2 Energy Quantization and Uncertainty Principle

Classical and quantum mechanics provide different descriptions of the same physical system. This is true even for the simplest cases. As an example, consider a harmonic oscillator with mass m and frequency ω , in a thermal state at temperature T . Classically, its phase-space distribution is given by the following Gaussian:

$$P_{HO}^{clas}(x, p) = \frac{\omega}{2\pi k_B T} e^{-\left(\frac{m\omega^2 x^2}{2k_B T} + \frac{p^2}{2mk_B T}\right)}. \quad (4.1)$$

Strictly speaking, phase-space distributions are not part of the quantum mechanical framework, as it is impossible to simultaneously determine the position and momentum of a system. Nevertheless, quasi-probability distributions, such as the Wigner function, share many properties with phase-space distributions [1, 2] and are considered the “closest quantum equivalent” to a phase-space distribution. The Wigner function for the same harmonic oscillator at the same state is

$$P_{HO}^{quan}(x, p) = \frac{\tanh[\frac{\hbar\omega}{2k_B T}]}{\hbar\pi} e^{-\tanh[\frac{\hbar\omega}{2k_B T}]\left(\frac{m\omega x^2}{\hbar} + \frac{p^2}{\hbar m\omega}\right)}. \quad (4.2)$$

These two distributions coincide at the regime where the thermal energy is much larger than the quantization energy, i.e., $\frac{\hbar\omega}{k_B T} \ll 1$. In this limit, the classical description is very precise, so quantum effects can be neglected. Nevertheless, the distributions diverge at low temperatures (see Fig. 4.1), where classical mechanics predicts a smaller and smaller position and momentum uncertainties, in contradiction with the uncertainty principle. In contrast, the Wigner distribution, Eq. (4.2), obeys the uncertainty principle at any temperature.

This is the main difference between quantum and classical distributions [2]. Moreover, confining potentials impose boundary conditions that quantize the energy state, create a zero point energy, and establish a dependence of the system energy on the boundary conditions. The combination of all these effects results on different energy, heat capacity and other thermal properties [3] relative to the ones predicted by classical mechanics and thermodynamics that neglect boundary effects [4]. In this section of the chapter we will explore how those differences affect the performance of thermal

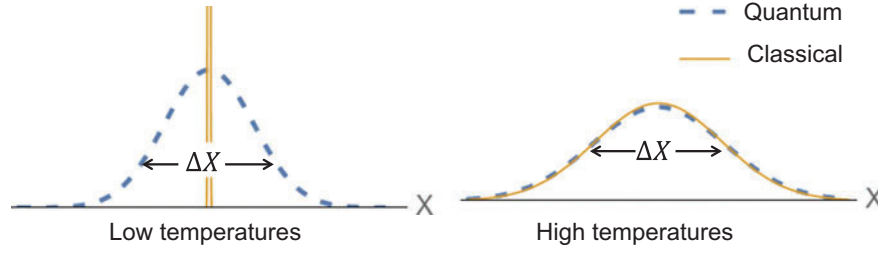


Fig. 4.1 Position projection of the classical (yellow continuous line) and quantum (blue dashed line) phase-space distributions for a harmonic oscillator in a thermal state. Both distributions coincide at high temperatures (right), but at low temperatures (left) the quantum Wigner distribution, in contrast with the classical one, has to comply with the uncertainty principle, forcing the distributions to diverge

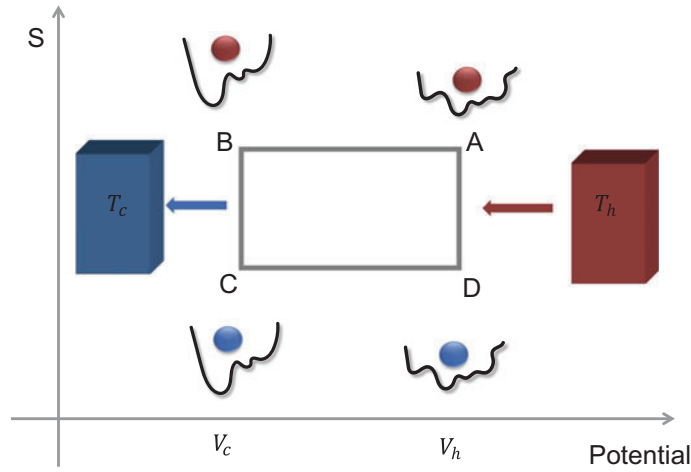


Fig. 4.2 Otto cycle. The x-axis represents a parameter that controls the working medium potential and the y-axis represents its Von Neumann entropy

heat machines. We will compare exactly the same heat machine, i.e., same working medium, same baths, same temperatures, etc., where in one case the behavior of the working medium is dictated by classical mechanics and in the other by quantum mechanics. In this section we do not consider the effects of quantum coherences or quantum correlations. These are left for subsequent sections of this chapter.

In particular, we consider an Otto heat machine [5–7] operating with a M -dimensional working medium, but a similar analysis can be done for other cycles. The Otto cycle is composed of the following four strokes (see Fig. 4.2): At point A of the cycle the working medium potential is $V_h(x_1, \dots, x_M) \equiv V_h$. Its Hamiltonian is H_h and its energy levels are $\{E_n^h\}$. The working medium is in thermal equilibrium with a hot bath at temperature T_h , so its state in the Hamiltonian energy basis is

$\rho_A = Z_{T_h}^{-1} \sum_n e^{-\frac{E_n^h}{k_B T_h}} |n\rangle\langle n|$, where Z_{T_h} is the partition function. At A , the adiabatic stroke starts: the working medium is decoupled from the hot bath and the potential is slowly deformed until point B , where it reaches $V_c(x_1, \dots, x_M) \equiv V_c$. Here, the Hamiltonian is H_c , and its energy levels are $\{E_n^c\}$. The deformation is slow enough for the process to fulfill the assumptions of the quantum adiabatic theorem [8], so the levels populations are the same $\rho_B = \rho_A$. At B , the system is coupled to the cold thermal bath at temperature T_c , initiating the cold isochoric stage. The system relaxes, achieving thermal equilibrium at C , i.e., $\rho_C = Z_{T_c}^{-1} \sum_n e^{-\frac{E_n^c}{k_B T_c}} |n\rangle\langle n|$, where Z_{T_c} is the partition function. Then, the working medium undergoes the second adiabatic stroke: it is decoupled from the bath and the potential is again slowly transformed back, reaching V_h at D . Here too, we apply the quantum adiabatic theorem so $\rho_D = \rho_C$. Finally, at this point, the second isochoric phase starts: the working medium is coupled again to the hot bath and equilibrates with it, returning to point A .

One useful feature of the Otto cycle is that work and heat are exchanged during different parts of the cycle: work is transferred during the adiabatic processes and heat over the isochoric strokes. The heat flow from the hot bath is

$$Q_h = \langle H_h \rangle_A - \langle H_h \rangle_D = \sum_n E_n^h \left(\frac{e^{-\frac{E_n^h}{k_B T_h}}}{Z_{T_h}} - \frac{e^{-\frac{E_n^c}{k_B T_c}}}{Z_{T_c}} \right) \quad (4.3)$$

and from the cold bath is

$$Q_c = \langle H_c \rangle_C - \langle H_c \rangle_B = \sum_n E_n^c \left(\frac{e^{-\frac{E_n^c}{k_B T_c}}}{Z_{T_c}} - \frac{e^{-\frac{E_n^h}{k_B T_h}}}{Z_{T_h}} \right). \quad (4.4)$$

Using the first law of thermodynamics, the expression for the work can be obtained,

$$W = -Q_h - Q_c = (E_n^c - E_n^h) \left(\frac{e^{-\frac{E_n^h}{k_B T_h}}}{Z_{T_h}} - \frac{e^{-\frac{E_n^c}{k_B T_c}}}{Z_{T_c}} \right). \quad (4.5)$$

Here, we use the sign convention that energy flowing to the working medium is positive and flowing from the working medium is negative. The heat machine has several operating modes: for $W < 0$ and $Q_h > 0$, it is a heat engine and its thermal efficiency quantifies the amount of work extracted per unit of incoming heat, i.e., $\eta = \frac{-W}{Q_h}$; for $W > 0$ and $Q_c > 0$, it is a refrigerator and its efficiency or coefficient of performance measures how much energy flows from the cold bath per unit of invested work $COP = \frac{Q_c}{W}$.

For a classical Otto heat machine, whose working medium is an ideal gas, the operation mode is determined by the compression ratio $r = \frac{Vol_c}{Vol_h}$, (Vol_c is the container volume at point C and Vol_h is the container volume at point A): for

$r > r_{Car} = \left(\frac{T_h}{T_c}\right)^{\frac{1}{\gamma-1}}$, it operates as a refrigerator and for $1 < r < r_{Car}$ the classical Otto heat machine operates as an engine with an efficiency

$$\eta = 1 - \frac{1}{r^{\gamma-1}} \leq 1 - \frac{1}{r_{Car}^{\gamma-1}} = \eta_{Car}, \quad (4.6)$$

where η_{Car} is the Carnot efficiency for an engine and $\gamma = \frac{C_p}{C_v}$ is the ratio of heat capacities. An important property of the classical Otto machine is that, in order to extract work or extract heat from the cold bath, the working medium has to be compressible [4, 9–12]. Otherwise, $r = 1$ and $\eta = 0$, precluding work extraction. This compression ratio is smaller than r_{Car} , hindering also the refrigerator operation. Here, quantum mechanics provides heat machines with an advantage over their classical counterparts. As it was shown on [5] and we explain below, once the working medium is governed by quantum laws instead of classical mechanics, compressibility is no longer required for work extraction or for cooling down the cold bath. Quantum mechanics allows the realization of heat machines with incompressible working media, opening the possibility for creating classically inconceivable heat machines.

One feature of a classical ideal gas undergoing an Otto cycle is that it is always at equilibrium, so at every point of the cycle it is possible to define the ideal gas temperature. This is not always true for a quantum working medium. For example,

at B , the working medium is at the state $\rho_B = Z_{T_h}^{-1} \sum_n e^{-\frac{E_n^h}{k_B T_h}} |n\rangle\langle n|$, which is a Boltzmann distribution of the energy levels at point A ($\{E_n^h\}$) instead of point B ($\{E_n^c\}$). So, in general, the state ρ_B is not a thermal equilibrium state with respect to the Hamiltonian at point B . The only exception is if the transformation homogeneously scales the energy levels i.e., $E_n^c = q E_n^h$, where q is independent of n . Under this condition we can rewrite the state at B as $\rho_B = Z_{qT_h}^{-1} \sum_n e^{-\frac{E_n^c}{k_B q T_h}} |n\rangle\langle n|$, which is a Boltzmann distribution at temperature qT_h of the energy levels at point B . In a similar way, the state at D , $\rho_D = Z_{T_c}^{-1} \sum_n e^{-\frac{E_n^c}{k_B T_c}} |n\rangle\langle n| = Z_{T_c/q}^{-1} \sum_n e^{-\frac{q E_n^h}{k_B T_c}} |n\rangle\langle n|$ is a Boltzmann distribution at temperature T_c/q of the energy levels at point D . Most of the research on quantum Otto cycles has focused on this type of transformations [6, 7] which includes the length change of a 1D infinite well [13], frequency shift of a 1D harmonic oscillator [14] or any other scale invariant transformation [15]. We will first study the differences between classical and quantum heat machines undergoing homogeneously energy scaling, which allows the working medium to be in a thermal state during the whole cycle. Then, we will explore the regime of inhomogeneous energy scaling, which has been seldom studied, with few exceptions [16, 17]. For an inhomogeneous energy scaling, the working medium deviates from a thermal state during the adiabatic strokes, resulting on more striking discrepancies between the operation of classical and quantum heat machines, and providing the later with more noteworthy advantages.

4.2.1 Homogeneous Energy Scaling: Work and Heat Corrections

In the case of the homogeneous energy scaling, the working medium is in a thermal state during the whole cycle. So we can use Wigner's original computation to calculate the quantum corrections to the performance of a heat machine. In 1932, Wigner calculated the quantum corrections to the quasiprobability distribution of an M -dimensional system in a potential V at thermal equilibrium at temperature T [2]. After integrating the momentum, the unnormalized position distribution is given by a series expansion on $\hbar$:

$$\int \dots \int dp_1 \dots dp_M P^{quan}(x_1, \dots, x_n; p_1, \dots, p_M) = P^{clas} + \hbar^2 P_2^{quan} + \hbar^4 P_4^{quan} + O(\hbar^6), \quad (4.7)$$

where

$$P^{clas} = e^{-\frac{V}{k_B T}} \quad (4.8)$$

is the classical phase-space distribution and P_{2n}^{quan} are quantum corrections that depend on the potential and its derivatives but not on $\hbar$. For example, the first quantum correction is

$$P_2^{quan} = e^{-\frac{V}{k_B T}} \left[-\frac{1}{12(k_B T)^2} \sum_k \frac{1}{m_k} \frac{\partial^2 V}{\partial x_k^2} + \frac{1}{24(k_B T)^3} \sum_k \frac{1}{m_k} \left(\frac{\partial V}{\partial x_k} \right)^2 \right]. \quad (4.9)$$

At high temperatures, the Wigner function tends to the classical phase space distribution, i.e., $\int \dots \int dp_1 \dots dp_M P^{quan}(x_1, \dots, x_n; p_1, \dots, p_M) \approx e^{-\frac{V}{k_B T}}$. But as the temperature decreases, quantum corrections need to be included to prevent the violation of the uncertainty principle. These corrections change the expectation value of the energy,

$$\langle H \rangle_T = E_{clas}(V, T) + \hbar^2 E_{2, QC}(V, T) + O(\hbar^4), \quad (4.10)$$

where $E_{clas}(V, T)$ corresponds to the system energy predicted by classical mechanics and $E_{2, QC}(V, T)$ is the first quantum correction

$$E_{2, QC}(V, T) = \frac{1}{\int P^{clas}} \left[\frac{1}{24k_B T} \int \sum_k \frac{1}{m_k} \frac{\partial^2 V}{\partial x_k^2} P^{clas} + \int V \times P_2^{quan} - \frac{\int V \times P^{clas} \int P_2^{quan}}{\int P^{clas}} \right], \quad (4.11)$$

where all the integrals are over all the positions, $x_1, \dots, x_M$. The quantum corrections to the energy affect the output of a heat machine. For example, the heat exchanged with the hot bath, including quantum corrections of the order $\hbar^2$ is

$$Q_h = Q_h^{clas} + \hbar^2 (E_{2, QC}(V_h, T_h) - E_{2, QC}(V_h, T_c/q)) + O(\hbar^4), \quad (4.12)$$

where Q_h^{clas} corresponds to the heat that a classical working medium would exchange with the hot bath during the Otto cycle, and the other terms are corrections that have to be included for a quantum working medium. In the same way, we can calculate the exchanged work,

$$W = W^{clas} - \hbar^2 (E_{2,QC}(V_h, T_h) - E_{2,QC}(V_h, T_c/q) + E_{2,QC}(V_c, T_c) - E_{2,QC}(V_c, qT_h)) + O(\hbar^4), \quad (4.13)$$

where W^{clas} corresponds to the work exchange by a classical Otto engine and the rest are the quantum corrections. As we explain later (see Eq. (4.17)), in order to extract work, i.e., $W < 0$, the scaling factor of the energy levels has to comply with the following inequality: $\frac{T_c}{T_h} < q < 1$.

For 1D systems in an Otto cycle with potentials $V_c = a_c x^{2n}$ and $V_h = a_h x^{2n}$, the scaling factor between energy levels is $q = \left(\frac{a_c}{a_h}\right)^{\frac{1}{1+n}}$ [7] and the correction of order $\hbar^2$ to the exchanged work can be analytically calculated. It is equal to

$$- \frac{\hbar^2 \pi (2n^2 + n - 1) \csc[\frac{\pi}{2n}]}{12m(\Gamma[\frac{1}{2n}])^2} \left(\frac{a_c}{k_B T_c}\right)^{\frac{1}{n}} \left(1 - \frac{1}{q}\right) \left(1 - \left(\frac{T_c}{qT_h}\right)^{\frac{1}{n}}\right), \quad (4.14)$$

where Γ is the gamma function. The correction can be positive or negative, but in the regime of work extraction, $W < 0$, it is always positive, i.e., it always reduces the work extraction.

The first quantum correction starts being relevant when the operation temperatures are not so high and any of the Wigner functions during the cycle deviate from the classical phase-space distribution. As temperatures decrease even more, higher order corrections should be considered. At least for 1D systems, there are recursive formulas that could be used to calculate any P_{2n}^{quan} [18], from which the full quantum corrections to the work and heat could be calculated. Nevertheless, this is not a practical approach and we will now introduce a simpler strategy that is exact at any temperature.

Using the fact that the energy levels are homogeneously scaled, $E_n^c = q E_n^h$, the expressions for the exchanged heat and work, Eqs. (4.3)–(4.5), can be rewritten. Consider for example Q_h (Eq. (4.3)). Using the relation between the energy levels this equation becomes

$$\begin{aligned} Q_h &= \sum_n E_n^h \left(\frac{e^{-\frac{E_n^h}{k_B T_h}}}{Z_{T_h}} - \frac{e^{-\frac{q E_n^h}{k_B T_c}}}{Z_{T_c/q}} \right) \\ &= \int_{T_c/q}^{T_h} \frac{d}{dT} \left(\sum_n E_n^h \frac{e^{-\frac{E_n^h}{k_B T}}}{Z_T} \right) dT = \int_{T_c/q}^{T_h} \frac{d\langle H_h \rangle_T}{dT} dT = \int_{T_c/q}^{T_h} C_v dT, \end{aligned} \quad (4.15)$$

where we have used the fundamental theorem of calculus and C_v is the heat capacity of a thermal state at temperature T with Hamiltonian H_h . In the same way, we can rewrite the equations for Q_c and W as:

$$Q_c = -q \int_{T_c/q}^{T_h} C_v dT, \quad (4.16)$$

$$W = (q - 1) \int_{T_c/q}^{T_h} C_v dT. \quad (4.17)$$

Because the heat capacity is always positive, work is extracted only if $\frac{T_c}{T_h} < q < 1$. We leave the readers to prove this as an exercise. All the information about the nature of the working medium is contained only on the heat capacity, C_v . If the working medium is a classical system, classical mechanics should be used to compute C_v . If the working medium is a quantum system, quantum mechanics should be used instead. Consider for example a working medium composed of N independent harmonic oscillators at thermal equilibrium at temperature T . A classical description, based on the equipartition theorem, assigns a contribution of $\frac{k_B}{2}$ for each quadratic degree of freedom of the Hamiltonian. There are two quadratic degrees of freedom for each harmonic oscillator, x and p . Therefore, classically $C_v^{HO,clas} = Nk_B$ at any T . In contrast, if we use quantum mechanics for calculating the heat capacity, we will get $C_v^{HO,quan} = Nk_B \left(\frac{\hbar\omega}{2k_B T} \text{csch} \left[\frac{\hbar\omega}{2k_B T} \right] \right)^2$, which is temperature dependent (see Fig. 4.3-left). Because, $C_v^{HO,quan} \leq C_v^{HO,clas}$, the heat and work exchanged by a heat machine made of a quantum harmonic oscillator working medium is always less than its classical counterpart, turning the classical heat machine into a better choice if we want to increase the heat or work exchange.

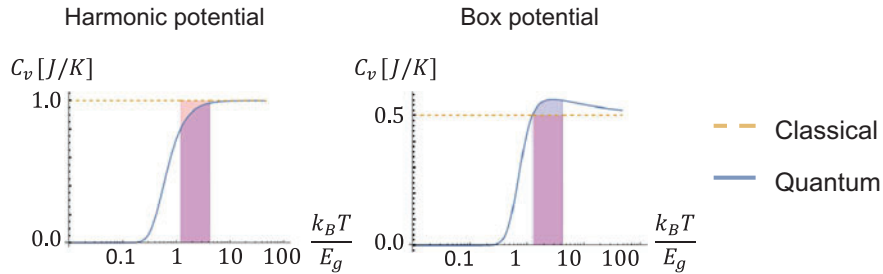


Fig. 4.3 Classical (yellow dashed line) and quantum (blue continuous line) heat capacities as function of the temperature normalized by the working medium ground state energy. The area below the curves, enclosed by the temperatures T_c/q and T_h , is proportional to the work and heat exchange by a classical or a quantum heat machine. For 1D harmonic oscillator potentials (left), the classical heat machine has a larger output than its quantum counterpart (see red area above the quantum curve). In contrast, for a particle in a 1D box (right), the output of the quantum heat machine is larger (see blue area above the classical curve)

Nevertheless, the advantage of the classical heat machine is not a general feature and depends on the type of working medium, which is determined by the potentials, and the temperature range. As an example consider a working medium composed of particles of mass m in a one dimensional box of length L . The quantum heat capacity is larger than its classical counterpart at not so low temperatures, i.e., $k_B T \gtrsim \frac{\hbar^2 \pi^2}{2mL^2}$ (see Fig. 4.3-right). Thus, for a one dimensional box potential, a quantum heat machine may have a larger output than its classical counterpart, i.e., extracts more work and exchanges more heat with the thermal baths. Which are the required features for a potential in order to boost the heat machine output and how they relate with other quantum effects, such as the Wigner function negativity, are still open questions that should be further investigated.

Up to this point we have talked about the work and heat exchange. But what about the thermal efficiencies, such as the engine efficiency or the coefficient of performance of a refrigerator? Because these quantities are rates of the exchanged work and the heat from one of the baths,

$$\eta = -\frac{W}{Q_h} = 1 - q; \quad COP = \frac{Q_c}{W} = \frac{q}{1 - q}, \quad (4.18)$$

they do not depend on the heat capacity, and therefore they do not depend on the quantum or classical nature of the working medium. Thus, $\eta^{quan} = \eta^{clas}$ and $COP^{quan} = COP^{clas}$. The efficiencies only depend on the potential deformation, which is characterized by the energy levels scaling factor, q .

4.2.2 Inhomogeneous Energy Scaling and the Efficiency Divergence

Could the efficiency of a quantum heat machine be greater than or at least different from its classical counterpart? Do quantum heat machines have any fundamental advantage over their classical counterparts? The answer to both questions is yes, but the potential transformation should be such that the energy levels are not homogeneously scaled, i.e., $E_n^c \neq q E_n^h$. As we show below, not only the performance is different, but energy quantization enables the realization of classically inconceivable heat machines, such as those operating with an incompressible working medium [5].

We exemplify these effects in a particular 2D example, but a similar analysis can be done for higher dimensions, such as 3D, or other potentials [5]. Here, we consider a working medium contained in a two dimensional box. At points D and A of the cycle, the dimensions of the box are L_x^h and L_y^h . At points B and C , they are L_x^c and L_y^c . If the working medium is incompressible, the box transformations during the adiabatic strokes have to keep the area constant: if the length along x is increased by a factor j , $L_x^c = j L_x^h$, then the size along y has to be scaled by a factor $\frac{1}{j}$, $L_y^c = \frac{1}{j} L_y^h$. These transformations correspond to the red lines on the plots on Fig. 4.4.

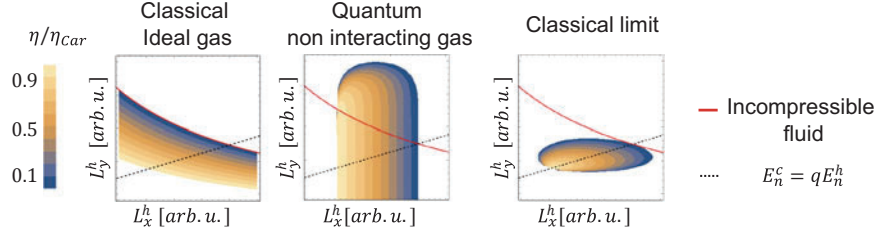


Fig. 4.4 Heat engine efficiency for a 2D potential box normalized by the Carnot efficiency. At point B and C of the cycle, the box lengths are such that $L_y^c/L_x^c = 0.25$ and the temperature ratio is $T_h/T_c = 2$. The plots show the results for different values of the box lengths at points D and A of the cycle, i.e., L_x^h and L_y^h . An incompressible working medium can only undergo area preserving transformations, here indicated by the red line. For these transformations there is no work extraction for a classical working medium, independently if it is an ideal gas (left) or the classical limit $\hbar \rightarrow 0$ (right) of a quantum particle in a box. In contrast, a quantum working medium (center) can be highly efficient, $\eta/\eta_{Car} > 0.9$, even for area preserving transformations. The dashed black line indicates transformations where the energy levels are homogeneously scaled and the quantum and classical efficiency coincide

During the adiabatic strokes, the adiabatic invariants have to be constant. The adiabatic invariant for a classical particle of mass m and energy E in a two dimensional box of area A is [19]

$$\mu = 2\pi mEA, \quad (4.19)$$

which implies that during the adiabatic transformation mEA is constant. So, for a constant area adiabatic transformation, the energy of the classical working medium does not change, precluding the work extraction. This is true for any classical non-interacting system, from an ideal gas to a single particle. Furthermore, the lack of work extraction is confirmed by taking the classical limit ($\hbar \rightarrow 0$) of the quantum calculation for the work, (see Eq. (4.5) and Fig. 4.4-center).

For a quantum working medium, one has to consider the energy levels, which are given by

$$E_{n_x, n_y}^i = \frac{\hbar^2 \pi^2}{2m} \left(\left(\frac{n_x}{L_x^i} \right)^2 + \left(\frac{n_y}{L_y^i} \right)^2 \right), \quad (4.20)$$

where n_x and n_y are integers, and $i = \{h, c\}$. For the constant area transformations, as long as $j \neq 1$, the energy levels scaling is inhomogeneous and the classical and quantum efficiencies differ. Moreover, Eq. (4.19) is not an adiabatic invariant at the quantum regime, where constant level populations is enough to warrant adiabaticity. As stated by the quantum adiabatic theorem, adiabaticity is achieved by performing the transformation slowly enough. This can be realized for constant area deformations that change the energy of the working medium. For simplicity, here we are assuming that the quantum working medium is composed of distinguishable particles.

In contrast to the classical heat machine, a quantum heat machine can be highly efficient, i.e., operate close to the Carnot limit, even for constant area transformations (see red line on Fig. 4.4-center). This opens the possibility of creating heat machines using incompressible working media, which classically would be impossible. Despite the increase in efficiency, quantum heat machines are always limited by the Carnot bound and fully comply with the second law of thermodynamics.

The quantum model of a particle in a two dimensional box is not the exact counterpart of a classical ideal gas heat machine. For an ideal gas, changes on one of the directions affect the other, redistributing the energy. In contrast, our quantum model is composed of two independent degrees of freedom that do not interact between themselves. Its classical limit, $\hbar \rightarrow 0$, corresponds to a different performance than the ideal gas heat machine (see Fig. 4.4 right and left, respectively). In any case, for a classical working medium contained in a two dimensional box, either an ideal gas or one composed of two independent degrees of freedom, classical adiabatic invariants forbid the work extraction for constant area transformations. This differs from the quantum case where quantum adiabatic invariants still allow work extraction. Further extension of this work should confirm if, for constant area transformations, work can be extracted using a “quantum ideal gas”. The gas Hamiltonian should include non-trivial interactions among its degrees of freedom, i.e., non-linear couplings, in order to avoid the formation of normal modes and to achieve full thermalization. Further works should also consider the effects of different quantum statistics, such as Fermi–Dirac or Bose–Einstein statistics.

In summary, in this section we have analyzed the differences between heat machines with a working medium governed either by classical or quantum mechanics. The difference in their operation is determined by the type of Hamiltonian transformation: if the energy levels are homogeneously scaled, the work and heat exchange may diverge, but the efficiencies are the same. In contrast, for inhomogeneous energy level scaling, the efficiencies also diverge, allowing the extraction of work with incompressible working fluids and opening for the realization of classically impossible heat machines. Future research should focus on finding other fundamental differences between the operation of classical and quantum heat machines. In this section, we have not studied quantum coherences or correlations. All the quantum effects that have been considered are the basic ones of a confined system: energy quantization and the uncertainty principle. Nevertheless, as we showed, this is enough for heat machines to have fundamentally different performances.

4.3 Coherence and Small Action Regime

Quantum coherence is one of the underlying principles of quantum mechanics that allows us to distinguish between classical and quantum phenomena. Historically, quantum coherence clarified the basic aspect of wave-particle duality in physical objects. Nowadays, scientists are trying to reveal the role of quantum coherence in biological, chemical, and physical systems and learn how to exploit it as a resource [20, 21].